

Update LOSO

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Note: updated table 09/12 can be found in 4. It shows the average marginal effect of different variables on the probability of receiving each type of advice. For continuous variables it is a 1 unit increase and for categorical is vs baseline (which is specified in parenthesis). The table is long, but it is intended to capture how adding different measures change the precision of the estimated marginal effects.

1 Introduction

In the past weeks I have checked what we discussed last time, here is the main summary of what I have done + results:

1. **Explain C-certificates (1) vs A (0) and explain B/C (failures) jointly vs $A \rightarrow C$** certificates are very noisy and hard to predict, but mostly capture bad performance. **It is hard to separate vs B certificate, as both capture almost the same in terms of performance, grades, student characteristics.** Difference between C and B certificate seem to be driven more by school characteristics(tracks offered, rules etc) than by performance. Check section 2.1.
2. **Jointly explain A/B/C with ordered logit** $\rightarrow$ needed to create a dataset with all schools and all school characteristics that LOSO provides, as I could not control for school FE. We have very rich characteristics of the supply side; I have added a brief description in section 4. Results of the ordered logit can be checked at section 2.2 and are similar as in point 1.
3. **Conditional on having received an A certificate, estimate a model of teacher advice** $\rightarrow$ it seems like teachers, conditional on absolute performance (std exams), effort and student characteristics are making recommendations based on class relative rankings. This may be due to school constraints. Further exploration of the supply side of schools in terms of subtracks is needed (I am working on that). Also, there is a striking gender pattern in advice. Results and interpretation can be checked in section 3.

2 A/B/C Certificates

2.1 Linear model C vs A, B/C vs A

C-certificates are much harder to predict than B-certificates (check table 8). In Grade 2 they are extremely rare (about 4%), so part of the variation simply reflects noise or shocks that we do not observe. Even with the full set of controls, only about 24% of the variance is explained. The variables that matter for C are very intuitive: low conscientiousness (likely capturing behavior and discipline), very low teacher-reported rankings in Dutch and math, and very low effort in class, especially in math. These are the students who are both low-effort and low-performance.

When we **combine B and C together (bad outcomes) versus A (clean pass)**, the picture **changes slightly** (check table 9). Now SES becomes clearly important: higher parental education protects against receiving B or C. This makes sense because B-certificates are given at the margin, where teachers form expectations about future progression, and higher SES may signal support and capacity to catch up. Track also matters strongly for B but not for C, suggesting that B-certificates reflect school- or track-specific policies, while C-certificates are more purely performance-driven. Effort in math remains a strong predictor in both cases, but standardized grades matter more for B than for C. As expected,

pooling B and C produces much higher R-squared values.

Finally, we can also compare in a self-selected sample of "fails", those that get a B vs those that get a C certificate (table 10). We can see that those that get B/C certificate don't differ much in terms of observables, as no coefficient is statistically significant. Differences are more on the type of track they were in at grade 2 and the school they were in: capacity effects? In ASO much more likely to offer B? If we introduce school characteristics instead of school FE: we see that ASO elite schools are more likely to make repeat vs multilateral or TSO/BSO schools. Also, short term schools (schools that only offer cycle 1) are more likely to give C certificates (they can't issue B's and let students go somewhere else?) Capacity also matters and previous grading rules (strictness of the institution).

2.2 Ordered logit for A/B/C certificates

I now estimate an ordered logit where I model jointly the probability of receiving an A-, B- or C-certificate in Grade 2. I treat the three outcomes as ordered categories, with

$$C < B < A,$$

so a positive coefficient means that the variable shifts probability mass towards better outcomes (more A, fewer B/C).

I cannot include school fixed effects here because of the incidental-parameters problem in nonlinear panel models. **Instead, in specification (2) I control for a rich set of school characteristics:** share of girls at baseline¹, share of migrant-background students at baseline, school type (ASO only, TSO/BSO only, multilateral (all tracks) and), cycles offered (1, 1+2, 1+2+3, 1+2+3+4)², the historical share of A and B certificates in 1991 (proxy for stricter or more lenient grading rules), school capacity (total number of students) and whether the school is gender-balanced and since when it is so. I also include an indicator for whether the class teacher formally guides study-choice decisions.

Results can be seen in table 10. Specification (1) includes only student-level variables (cognitive and non-cognitive skills, standardized grades, teacher-assessed ranks and effort, parental involvement and track). Specification (2) adds the school-level controls.

The coefficients show that teacher-reported Dutch and math rankings are by far the strongest predictors of certificate outcomes, followed by math effort and conscientiousness. When I add school controls, the effect of parental education becomes much weaker, suggesting that part of the SES gradient in the simpler models was driven by school sorting. School composition (share migrants), school type and whether the class teacher guides the process also shift the distribution away from C/B and towards A-certificates. **Overall, the ordered logit confirms the earlier binary regressions but puts them into a single joint model for A/B/C.**

The second cutpoint (separating A from B/C) is well identified, but the first cutpoint (separating C from B/A) is imprecise. This pattern fits the data: we see a clear separation between A and the other outcomes, but relatively little information to pin down the exact boundary between B and the rare, noisy C-certificates. It can also be that X already do a good job so that the nuisance parameter does not add anything.

Table 1: Determinants of A/B/C Certificate Decisions (Ordered Logit)

	(1) Baseline	(2) + School Controls
Student characteristics		
Male (ref = female)	-0.152 (0.142)	-0.012 (0.112)
Numeric IQ (std.)	0.008 (0.006)	0.003 (0.008)
Verbal IQ (std.)	0.005 (0.006)	0.003 (0.007)
Conscientiousness (std.)	0.172* (0.090)	0.370*** (0.107)
Agreeableness (std.)	-0.010 (0.058)	-0.015 (0.073)
Extraversion (std.)	0.010 (0.071)	-0.120 (0.081)
Parental education (ref = low)		

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¹That is, using information on other cohorts.

²Cycle 1 is grade 1 and 2 only; cycle 2 is when formal tracking starts...

	(1) Baseline	(2) + School Controls
Medium parental education	0.340** (0.144)	0.124 (0.208)
High parental education	0.525*** (0.184)	0.077 (0.197)
Standardized grades		
Standardized Dutch grade	0.003 (0.073)	0.037 (0.071)
Standardized math grade	0.078** (0.035)	0.070* (0.042)
Teacher-reported math rank (ref = Q1)		
Q2	0.412*** (0.111)	0.334** (0.155)
Q3	1.014*** (0.191)	1.024*** (0.220)
Q4	1.373*** (0.260)	1.424*** (0.282)
Q5	2.829*** (0.568)	2.963*** (0.681)
Teacher-reported Dutch rank (ref = Q1)		
Q2	0.393*** (0.152)	0.606*** (0.162)
Q3	1.178*** (0.158)	1.498*** (0.183)
Q4	2.138*** (0.338)	2.342*** (0.403)
Q5	2.027*** (0.444)	2.308*** (0.539)
Subject-specific effort (ref = level 1)		
Math effort level 2	0.702*** (0.172)	0.658*** (0.169)
Math effort level 3	1.260*** (0.192)	1.211*** (0.221)
Math effort level 4	1.995*** (0.473)	1.769*** (0.479)
Dutch effort level 2	0.111 (0.160)	-0.020 (0.199)
Dutch effort level 3	0.071 (0.211)	-0.095 (0.271)
Dutch effort level 4	-0.143 (0.268)	-0.362 (0.309)
Parental involvement (std.)	0.095 (0.067)	0.150* (0.085)
Track (ref = BSO)		
TSO	-1.783*** (0.395)	-2.143*** (0.450)
ASO	-1.563*** (0.348)	-1.815*** (0.499)
School characteristics		
Share girls (%)		-0.000 (0.009)
Share migrants (%)		-0.028** (0.014)
School type (ref=ASO-only)		
School type: autms		-0.020 (0.370)
School type: mltal		-0.048 (0.335)
School type: tso/bsu only		1.022* (0.590)
Cycles Offered (ref=cycle 1)		
Cycle: 1+2 offered		1.713** (0.704)
Cycle: 1+2+3 offered		0.070 (0.397)
Cycle: 1+2+3+4 offered		0.732 (0.762)
Historical A-share (1991)		0.042 (0.027)
Historical B-share (1991)		0.005 (0.027)
Capacity (school size)		-0.001 (0.001)
Gender-balanced school		0.027 (0.017)
Class teacher guides choice		1.435*** (0.508)
Cutpoints		
Cut 1	-0.430 (0.898)	1.688 (2.873)
Cut 2	2.193*** (0.741)	4.755* (2.677)
Observations	3014	2029

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3 Teacher advice

Conditional on having received an A-certificate at grade 2, I estimate a multinomial logit on the type of advice students receive from grade 2 to 3 in the choice set $J = \{\text{ASOstem}, \text{ASOnstem}, \text{TSOstem}, \text{TSOnstem}, \text{BSO}\}$.

Conditional on an A-certificate, teacher recommendations show systematic patterns across cognitive,

behavioral, socioeconomic, and institutional dimensions. Performance indicators—especially teacher-based ranks in math and Dutch—are by far the strongest predictors of advice, sharply increasing the likelihood of ASO and STEM recommendations and reducing the probability of BSO. Conscientiousness also matters, shifting students toward academic tracks and away from vocational ones. High-SES students are disproportionately steered into ASO–STEM, but SES effects disappear for other outcomes once performance and school context are controlled (this suggests SES sorting into schools at grade 1). Also, the direction of advice matches the strength in relative rankings per subject: strength in math fosters STEM recommendation and relative strength in dutch fosters NonStem recommendation.

Even after conditioning on IQ, standardized exam grades, and effort (one latent factor of ability/interest), teachers still recommend STEM vs. non-STEM based on students' relative rankings within their class (is this desirable?). One possible interpretation is that teachers are constrained by school capacity: starting from grade 3, schools may be forced to split classes into STEM and non-STEM groups because there is not enough capacity to place all students in certain programs (e.g., not enough STEM teachers). This would push teachers to allocate students based on relative rather than absolute skill³. But even without capacity constraints, teachers may rely on local, norm-based comparisons when forming judgments, using relative rather than absolute performance as the relevant benchmark—so peer composition becomes key. More information on the supply side of tracks and subtracks is needed to say something more concrete here (working on it!!).

Also, even when we control for performance, ability, and other characteristics, gender differences persist, with boys more likely to receive STEM-oriented advice, particularly within TSO⁴. Effort measures, which strongly influence A/B/C certificates and track advice, play essentially no role in horizontal guidance.

Finally, school characteristics impose binding constraints: students in TSO/BSO-oriented schools almost never receive ASO advice, and longer-cycle schools are more likely to channel students toward ASO–STEM.

³Currently I am trying to recover supply side information on the subtracks available in each school.

⁴It is hard to rule out that teachers recommend what students already want and that there is a negotiation process throughout the year. Controlling for beliefs at the beginning of grade 2 about the track-subtrack students expect to graduate from may help distinguish this.

Table 2: Marginal Effects on Teacher Track–Subtrack Advice (Given A Certificate)

	ASO-STEM (1)	ASO-nonSTEM (2)	TSO-STEM (3)	TSO-nonSTEM (4)	BSO (5)
Personality Traits					
Conscientiousness	0.078*** (0.017)	0.005 (0.016)	0.044*** (0.011)	0.018 (0.011)	-0.145*** (0.009)
Agreeableness	-0.004 (0.010)	0.004 (0.010)	-0.011 (0.008)	0.001 (0.008)	0.009 (0.008)
Extraversion	0.005 (0.010)	-0.017 (0.011)	-0.008 (0.008)	-0.008 (0.009)	0.029*** (0.008)
Gender & SES					
Male	0.051*** (0.019)	-0.065*** (0.020)	0.138*** (0.024)	-0.080*** (0.025)	-0.043* (0.023)
High SES	0.038** (0.017)	-0.010 (0.019)	0.006 (0.019)	-0.020 (0.019)	-0.015 (0.020)
Grades					
Math grade (std.)	0.039*** (0.011)	-0.014 (0.009)	0.001 (0.006)	-0.009 (0.006)	-0.017*** (0.005)
Dutch grade (std.)	-0.010 (0.012)	0.038*** (0.014)	-0.019** (0.008)	0.010 (0.011)	-0.019** (0.008)
Inertia (previous track)					
Worse track (lag)	-0.171*** (0.028)	-0.094*** (0.024)	0.129*** (0.024)	0.063*** (0.023)	0.073*** (0.020)
Better track (lag)	0.004 (0.036)	0.019 (0.038)	-0.078 (0.063)	-0.054 (0.035)	0.109* (0.058)
Effort (Dutch)					
Effort Q2	0.028 (0.045)	-0.061 (0.052)	0.057* (0.034)	0.012 (0.037)	-0.036 (0.038)
Effort Q3	0.002 (0.042)	-0.040 (0.051)	0.061* (0.034)	0.032 (0.037)	-0.056 (0.037)
Effort Q4	0.008 (0.043)	-0.047 (0.052)	0.015 (0.037)	0.050 (0.042)	-0.025 (0.042)
Effort (Math)					
Effort Q2	0.053 (0.064)	-0.040 (0.066)	0.027 (0.040)	0.007 (0.043)	-0.047 (0.044)
Effort Q3	0.084 (0.061)	-0.035 (0.065)	0.029 (0.041)	-0.015 (0.044)	-0.064 (0.046)
Effort Q4	0.091 (0.062)	-0.037 (0.067)	0.089* (0.047)	-0.032 (0.048)	-0.111** (0.049)
Class Rank (Math)					
Rank Q2	0.057 (0.057)	0.051 (0.061)	0.048 (0.036)	-0.055 (0.042)	-0.045* (0.026)
Rank Q3	0.137*** (0.053)	-0.036 (0.057)	0.012 (0.036)	-0.106** (0.042)	-0.007 (0.027)
Rank Q4	0.260*** (0.054)	-0.173*** (0.058)	-0.008 (0.039)	-0.136*** (0.044)	0.057* (0.031)
Rank Q5 (best)	0.279*** (0.056)	-0.200*** (0.060)	-0.035 (0.040)	-0.171*** (0.045)	0.127*** (0.035)
Class Rank (Dutch)					
Rank Q2	-0.077 (0.053)	0.126*** (0.046)	-0.036 (0.038)	0.041 (0.039)	-0.053 (0.034)
Rank Q3	0.002 (0.047)	0.116*** (0.040)	-0.055 (0.037)	0.001 (0.037)	-0.063* (0.035)
Rank Q4	0.038 (0.048)	0.094** (0.041)	-0.020 (0.039)	-0.068* (0.038)	-0.044 (0.036)
Rank Q5	0.060 (0.049)	0.039 (0.042)	-0.028 (0.041)	-0.031 (0.041)	-0.039 (0.039)
School Characteristics					
Share girls	0.000 (0.001)	-0.000 (0.001)	-0.001 (0.000)	0.000 (0.000)	0.001 (0.000)
Share migrants	0.002 (0.001)	0.003** (0.001)	-0.003*** (0.001)	-0.002*** (0.001)	0.001 (0.001)
School Type (ref = ASO-only)					
AUT/MS	0.022 (0.037)	-0.178*** (0.035)	0.078** (0.036)	-0.040 (0.025)	0.118*** (0.033)
MLT/ATL	-0.044 (0.034)	-0.133*** (0.034)	0.091*** (0.025)	0.009 (0.025)	0.077*** (0.027)
TSO/BSO only	-0.396*** (0.021)	-0.301*** (0.036)	0.351*** (0.048)	0.181*** (0.053)	0.165*** (0.035)
Observations	1609				

Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 3: Marginal Effects on Teacher Track–Subtrack Advice (Given A Certificate)
(Updated: Rank Quintiles for Dutch and Math)

	ASO-STEM (1)	ASO-nonSTEM (2)	TSO-STEM (3)	TSO-nonSTEM (4)	BSO (5)
Personality Traits					
Conscientiousness	0.247*** (0.012)	0.029*** (0.011)	-0.030*** (0.006)	-0.053*** (0.007)	-0.193*** (0.006)
Agreeableness	-0.010 (0.010)	0.000 (0.010)	-0.002 (0.006)	0.002 (0.007)	0.010* (0.006)
Extraversion	-0.035*** (0.009)	0.002 (0.009)	0.002 (0.006)	0.002 (0.008)	0.029*** (0.007)
Gender & SES					
Male	0.039*** (0.015)	-0.097*** (0.015)	0.202*** (0.012)	-0.122*** (0.013)	-0.021* (0.012)
High SES	0.132*** (0.017)	0.049*** (0.018)	-0.047*** (0.013)	-0.047*** (0.016)	-0.088*** (0.015)
Dutch Rank Quintiles (ref = Q1)					
Rank Q2	-0.019 (0.036)	0.002 (0.032)	-0.022 (0.020)	0.071*** (0.025)	-0.031* (0.016)
Rank Q3	0.048 (0.034)	0.045 (0.031)	-0.043** (0.021)	0.029 (0.026)	-0.079*** (0.019)
Rank Q4	0.139*** (0.037)	0.087** (0.036)	-0.023 (0.028)	-0.056* (0.030)	-0.147*** (0.027)
Rank Q5	0.139*** (0.041)	0.163*** (0.047)	-0.109*** (0.027)	-0.156*** (0.025)	-0.037 (0.047)
Math Rank Quintiles (ref = Q1)					
Rank Q2	0.058* (0.035)	-0.046 (0.035)	-0.031 (0.019)	0.042** (0.021)	-0.023 (0.018)
Rank Q3	0.112*** (0.034)	-0.000 (0.035)	0.011 (0.021)	0.023 (0.022)	-0.146*** (0.019)
Rank Q4	0.189*** (0.035)	0.003 (0.037)	-0.014 (0.025)	-0.016 (0.025)	-0.162*** (0.024)
Rank Q5	0.270*** (0.039)	-0.072* (0.040)	-0.032 (0.032)	-0.059* (0.034)	-0.108** (0.045)
Inertia (previous track)					
Worse track (lag)	-0.210*** (0.024)	-0.057** (0.025)	0.104*** (0.018)	0.093*** (0.018)	0.070*** (0.015)
Better track (lag)	-0.042 (0.041)	0.195*** (0.049)	-0.104*** (0.020)	-0.050 (0.033)	0.001 (0.035)
Effort (Dutch)					
Effort Q2	0.005 (0.043)	-0.018 (0.047)	0.041 (0.029)	-0.003 (0.034)	-0.026 (0.029)
Effort Q3	-0.009 (0.041)	-0.014 (0.045)	0.034 (0.028)	0.034 (0.035)	-0.045 (0.029)
Effort Q4	-0.028 (0.041)	0.014 (0.047)	0.020 (0.030)	0.029 (0.038)	-0.036 (0.032)
Effort (Math)					
Effort Q2	-0.029 (0.061)	0.054 (0.056)	0.028 (0.030)	-0.025 (0.044)	-0.028 (0.034)
Effort Q3	0.075 (0.059)	0.029 (0.054)	0.036 (0.031)	-0.078* (0.045)	-0.063* (0.035)
Effort Q4	0.071 (0.059)	0.010 (0.056)	0.091** (0.036)	-0.088* (0.048)	-0.084** (0.038)
School Characteristics					
Share girls	0.000	-0.000	-0.001	0.000	0.001
Share migrants	0.002	0.004***	-0.003***	-0.002***	0.000
School Type (ref = ASO-only)					
AUT/MS	0.018	-0.198***	0.081**	-0.027	0.126***
MLT/LAT	-0.042	-0.157***	0.084***	0.021	0.094**
TSO/BSO only	-0.396***	-0.315***	0.410***	0.145***	0.157***
Observations	1609				

Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Update: It may be that class ranking in performance captures the nonlinearities in ability that teachers use to make advice. To avoid this, I will control for ranking in the overall distribution of standardized exams in math and dutch. Results can be found in table 4⁵.

⁵Numeric and Verbal Iq are also included in the base specification. Sorry for not including it

Table 4: Determinants of teacher advice, conditional on A certificate

	(1) Base	+ std Grades	+ In	+ Effort/Perf	+ School Ch
c_CON					
ASO_stem	0.247*** (0.012)	0.130*** (0.015)	0.116*** (0.015)	0.091*** (0.015)	0.067*** (0.017)
ASO_nstem	0.029*** (0.011)	0.002 (0.015)	-0.006 (0.015)	0.017 (0.015)	-0.009 (0.017)
TSO_stem	-0.030*** (0.006)	-0.011 (0.008)	-0.004 (0.008)	-0.013 (0.009)	0.033*** (0.011)
TSO_nstem	-0.053*** (0.007)	-0.001 (0.009)	0.009 (0.009)	0.012 (0.009)	0.025** (0.011)
BSO	-0.193*** (0.006)	-0.121*** (0.007)	-0.116*** (0.007)	-0.108*** (0.007)	-0.116*** (0.010)
c_AGREE					
ASO_stem	-0.010 (0.010)	-0.011 (0.010)	-0.012 (0.009)	-0.023** (0.010)	-0.001 (0.010)
ASO_nstem	0.000 (0.010)	0.010 (0.010)	0.009 (0.010)	0.016 (0.010)	0.005 (0.010)
TSO_stem	-0.002 (0.006)	0.001 (0.006)	0.003 (0.006)	0.004 (0.006)	-0.013 (0.008)
TSO_nstem	0.002 (0.007)	-0.005 (0.007)	-0.004 (0.007)	-0.004 (0.007)	0.002 (0.008)
BSO	0.010* (0.006)	0.005 (0.006)	0.005 (0.006)	0.007 (0.006)	0.006 (0.007)
c_EXTRA					
ASO_stem	-0.035*** (0.009)	-0.009 (0.009)	-0.011 (0.009)	-0.002 (0.009)	0.006 (0.010)
ASO_nstem	0.002 (0.009)	0.000 (0.010)	-0.001 (0.010)	-0.009 (0.010)	-0.015 (0.011)
TSO_stem	0.002 (0.006)	-0.004 (0.007)	-0.003 (0.006)	-0.000 (0.007)	-0.007 (0.008)
TSO_nstem	0.002 (0.008)	-0.006 (0.008)	-0.005 (0.008)	-0.010 (0.008)	-0.008 (0.009)
BSO	0.029*** (0.007)	0.019*** (0.006)	0.019*** (0.006)	0.022*** (0.006)	0.024*** (0.008)
Male					
ASO_stem	0.039*** (0.015)	0.069*** (0.015)	0.072*** (0.015)	0.049*** (0.015)	0.050*** (0.019)
ASO_nstem	-0.097*** (0.015)	-0.088*** (0.016)	-0.085*** (0.016)	-0.066*** (0.016)	-0.062*** (0.020)
TSO_stem	0.202*** (0.012)	0.204*** (0.013)	0.200*** (0.012)	0.211*** (0.013)	0.161*** (0.023)
TSO_nstem	-0.122*** (0.013)	-0.133*** (0.013)	-0.136*** (0.013)	-0.129*** (0.013)	-0.088*** (0.025)
BSO	-0.021* (0.012)	-0.052*** (0.012)	-0.051*** (0.012)	-0.065*** (0.012)	-0.061*** (0.023)
high_ses					
ASO_stem	0.132*** (0.017)	0.076*** (0.017)	0.061*** (0.016)	0.045*** (0.016)	0.034** (0.017)
ASO_nstem	0.049*** (0.018)	0.026 (0.018)	0.023 (0.018)	0.016 (0.017)	-0.017 (0.019)

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Determinants of teacher advice, conditional on A certificate (continued)

	(1) Base	+ std Grades	+ In	+ Effort/Perf	+ School Ch
TSO_stem	-0.047*** (0.013)	-0.039*** (0.013)	-0.028** (0.014)	-0.020 (0.015)	0.008 (0.019)
TSO_nstem	-0.047*** (0.016)	-0.016 (0.017)	-0.011 (0.017)	0.002 (0.017)	-0.017 (0.019)
BSO	-0.088*** (0.015)	-0.047*** (0.016)	-0.045*** (0.016)	-0.042*** (0.016)	-0.008 (0.020)
2.rank_std_dutch (vs1)					
ASO_stem		-0.019 (0.036)	0.003 (0.035)	0.034 (0.037)	0.012 (0.036)
ASO_nstem		0.002 (0.032)	0.003 (0.032)	-0.015 (0.031)	-0.022 (0.032)
TSO_stem		-0.022 (0.020)	-0.024 (0.019)	-0.039* (0.021)	-0.033 (0.023)
TSO_nstem		0.071*** (0.025)	0.056** (0.024)	0.054** (0.022)	0.062*** (0.022)
BSO		-0.031* (0.016)	-0.038** (0.016)	-0.034** (0.017)	-0.020 (0.019)
3.rank_std_dutch (vs1)					
ASO_stem		0.048 (0.034)	0.057* (0.033)	0.065* (0.034)	-0.003 (0.035)
ASO_nstem		0.045 (0.031)	0.036 (0.031)	0.022 (0.031)	0.017 (0.033)
TSO_stem		-0.043** (0.021)	-0.043** (0.020)	-0.057** (0.022)	-0.032 (0.025)
TSO_nstem		0.029 (0.026)	0.029 (0.025)	0.049** (0.024)	0.076*** (0.025)
BSO		-0.079*** (0.019)	-0.079*** (0.019)	-0.079*** (0.020)	-0.057** (0.023)
4.rank_std_dutch (vs1)					
ASO_stem		0.139*** (0.037)	0.129*** (0.035)	0.101*** (0.036)	0.028 (0.037)
ASO_nstem		0.087** (0.036)	0.076** (0.036)	0.086** (0.036)	0.075* (0.039)
TSO_stem		-0.023 (0.028)	-0.009 (0.027)	-0.027 (0.030)	0.048 (0.034)
TSO_nstem		-0.056* (0.030)	-0.043 (0.029)	-0.005 (0.029)	0.005 (0.030)
BSO		-0.147*** (0.027)	-0.153*** (0.027)	-0.154*** (0.027)	-0.155*** (0.030)
5.rank_std_dutch (vs1)					
ASO_stem		0.139*** (0.041)	0.109*** (0.039)	0.054 (0.039)	-0.022 (0.039)
ASO_nstem		0.163*** (0.047)	0.143*** (0.045)	0.195*** (0.045)	0.120*** (0.046)
TSO_stem		-0.109*** (0.027)	-0.086*** (0.030)	-0.105*** (0.031)	-0.050 (0.046)
TSO_nstem		-0.156*** (0.025)	-0.148*** (0.026)	-0.105*** (0.028)	-0.057 (0.035)
BSO		-0.037 (0.047)	-0.018 (0.048)	-0.039 (0.045)	0.009 (0.062)
2.rank_std_math (vs1)					
ASO_stem		0.058* (0.035)	0.059* (0.035)	0.054 (0.038)	0.058 (0.040)
ASO_nstem		-0.046 (0.035)	-0.041 (0.035)	-0.046 (0.035)	-0.041 (0.035)

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Determinants of teacher advice, conditional on A certificate (continued)

	(1) Base	+ std Grades	+ In	+ Effort/Perf	+ School Ch
TSO_stem		-0.031 (0.019)	-0.030 (0.019)	-0.023 (0.020)	-0.036 (0.024)
TSO_nstem		0.042** (0.021)	0.041* (0.021)	0.056*** (0.021)	0.065*** (0.023)
BSO		-0.023 (0.018)	-0.029* (0.017)	-0.041** (0.019)	-0.047** (0.023)
3.rank_std_math (vs1)					
ASO_stem		0.112*** (0.034)	0.115*** (0.033)	0.105*** (0.036)	0.070* (0.039)
ASO_nstem		-0.000 (0.035)	-0.001 (0.035)	0.002 (0.035)	0.005 (0.035)
TSO_stem		0.011 (0.021)	0.012 (0.021)	0.011 (0.023)	0.036 (0.027)
TSO_nstem		0.023 (0.022)	0.024 (0.022)	0.031 (0.022)	0.025 (0.024)
BSO		-0.146*** (0.019)	-0.150*** (0.019)	-0.150*** (0.021)	-0.136*** (0.026)
4.rank_std_math (vs1)					
ASO_stem		0.189*** (0.035)	0.194*** (0.035)	0.166*** (0.038)	0.121*** (0.041)
ASO_nstem		0.003 (0.037)	0.004 (0.037)	0.016 (0.037)	0.032 (0.039)
TSO_stem		-0.014 (0.025)	-0.014 (0.025)	-0.002 (0.027)	0.029 (0.033)
TSO_nstem		-0.016 (0.025)	-0.019 (0.025)	-0.018 (0.024)	0.008 (0.028)
BSO		-0.162*** (0.024)	-0.165*** (0.024)	-0.162*** (0.026)	-0.190*** (0.032)
5.rank_std_math (vs1)					
ASO_stem		0.270*** (0.039)	0.270*** (0.039)	0.217*** (0.042)	0.169*** (0.045)
ASO_nstem		-0.072* (0.040)	-0.069* (0.039)	-0.010 (0.042)	0.007 (0.046)
TSO_stem		-0.032 (0.032)	-0.037 (0.031)	-0.039 (0.033)	0.009 (0.042)
TSO_nstem		-0.059* (0.034)	-0.057 (0.035)	-0.033 (0.035)	-0.019 (0.043)
BSO		-0.108** (0.045)	-0.108** (0.046)	-0.135*** (0.045)	-0.165*** (0.055)
1.worse_track					
ASO_stem			-0.210*** (0.024)	-0.210*** (0.027)	-0.173*** (0.028)
ASO_nstem			-0.057** (0.025)	-0.072*** (0.025)	-0.084*** (0.025)
TSO_stem			0.104*** (0.018)	0.129*** (0.021)	0.141*** (0.026)
TSO_nstem			0.093*** (0.018)	0.091*** (0.019)	0.044** (0.021)
BSO			0.070*** (0.015)	0.063*** (0.016)	0.072*** (0.019)
1.better_track					
ASO_stem			-0.042 (0.041)	-0.058 (0.039)	-0.007 (0.035)
ASO_nstem			0.195*** (0.049)	0.178*** (0.046)	0.014 (0.037)

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Determinants of teacher advice, conditional on A certificate (continued)

	(1) Base	+ std Grades	+ In	+ Effort/Perf	+ School Ch
TSO_stem			-0.104*** (0.020)	-0.109*** (0.022)	-0.094* (0.055)
TSO_nstem			-0.050 (0.033)	-0.082*** (0.028)	-0.072** (0.028)
BSO			0.001 (0.035)	0.071* (0.037)	0.159*** (0.053)
2.rank_tch_math(vs1)					
ASO_stem				0.054 (0.045)	0.001 (0.057)
ASO_nstem				-0.025 (0.047)	0.039 (0.060)
TSO_stem				0.042 (0.027)	0.049 (0.036)
TSO_nstem				-0.017 (0.026)	-0.050 (0.037)
BSO				-0.054*** (0.019)	-0.039 (0.024)
3.rank_tch_math (vs1)					
ASO_stem				0.159*** (0.041)	0.141*** (0.054)
ASO_nstem				-0.102** (0.044)	-0.052 (0.057)
TSO_stem				0.015 (0.027)	0.005 (0.035)
TSO_nstem				-0.051* (0.027)	-0.096*** (0.037)
BSO				-0.020 (0.020)	0.002 (0.026)
4.rank_tch_math (vs1)					
ASO_stem				0.265*** (0.041)	0.256*** (0.054)
ASO_nstem				-0.207*** (0.045)	-0.198*** (0.058)
TSO_stem				-0.027 (0.028)	-0.019 (0.038)
TSO_nstem				-0.082*** (0.029)	-0.113*** (0.040)
BSO				0.052** (0.024)	0.074** (0.030)
5.rank_tch_math (vs1)					
ASO_stem				0.281*** (0.043)	0.267*** (0.057)
ASO_nstem				-0.296*** (0.046)	-0.224*** (0.059)
TSO_stem				-0.009 (0.031)	-0.028 (0.040)
TSO_nstem				-0.067** (0.033)	-0.139*** (0.042)
BSO				0.091*** (0.027)	0.124*** (0.033)
2.rank_tch_dutch (vs1)					
ASO_stem				-0.040 (0.053)	-0.075 (0.054)
ASO_nstem				0.092* (0.048)	0.141*** (0.046)

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Determinants of teacher advice, conditional on A certificate (continued)

	(1) Base	+ std Grades	+ In	+ Effort/Perf	+ School Ch
TSO_stem				-0.022 (0.027)	-0.037 (0.037)
TSO_nstem				0.031 (0.030)	0.031 (0.036)
BSO				-0.062*** (0.024)	-0.060* (0.032)
3.rank_tch_dutch (vs1)					
ASO_stem				0.068 (0.048)	-0.003 (0.047)
ASO_nstem				0.007 (0.043)	0.131*** (0.039)
TSO_stem				-0.024 (0.027)	-0.057 (0.036)
TSO_nstem				0.015 (0.029)	0.000 (0.035)
BSO				-0.066*** (0.024)	-0.071** (0.032)
4.rank_tch_dutch (vs1)					
ASO_stem				0.076 (0.048)	0.036 (0.048)
ASO_nstem				0.010 (0.044)	0.095** (0.040)
TSO_stem				0.004 (0.029)	-0.018 (0.038)
TSO_nstem				-0.065** (0.029)	-0.063* (0.035)
BSO				-0.025 (0.026)	-0.050 (0.035)
5.rank_tch_dutch (vs1)					
ASO_stem				0.070 (0.050)	0.050 (0.049)
ASO_nstem				-0.042 (0.045)	0.033 (0.041)
TSO_stem				0.017 (0.031)	-0.027 (0.041)
TSO_nstem				-0.039 (0.033)	-0.019 (0.040)
BSO				-0.006 (0.028)	-0.037 (0.037)
2.rank_effort_dutch (vs1)					
ASO_stem				0.005 (0.043)	0.024 (0.045)
ASO_nstem				-0.018 (0.047)	-0.055 (0.050)
TSO_stem				0.041 (0.029)	0.039 (0.036)
TSO_nstem				-0.003 (0.034)	0.001 (0.038)
BSO				-0.026 (0.029)	-0.010 (0.034)
3.rank_effort_dutch (vs1)					
ASO_stem				-0.009 (0.041)	-0.001 (0.043)
ASO_nstem				-0.014 (0.045)	-0.022 (0.048)

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Determinants of teacher advice, conditional on A certificate (continued)

	(1) Base	+ std Grades	+ In	+ Effort/Perf	+ School Ch
TSO_stem				0.034 (0.028)	0.042 (0.036)
TSO_nstem				0.034 (0.035)	0.020 (0.038)
BSO				-0.045 (0.029)	-0.038 (0.033)
4.rank_effort_dutch (vs1)					
ASO_stem				-0.028 (0.041)	0.002 (0.043)
ASO_nstem				0.014 (0.047)	-0.032 (0.050)
TSO_stem				0.020 (0.030)	-0.000 (0.039)
TSO_nstem				0.029 (0.038)	0.037 (0.043)
BSO				-0.036 (0.032)	-0.007 (0.038)
2.rank_effort_math (vs1)					
ASO_stem				-0.029 (0.061)	0.063 (0.065)
ASO_nstem				0.054 (0.056)	-0.033 (0.065)
TSO_stem				0.028 (0.030)	0.022 (0.041)
TSO_nstem				-0.025 (0.044)	0.004 (0.045)
BSO				-0.028 (0.034)	-0.056 (0.043)
3.rank_effort_math (vs1)					
ASO_stem				0.075 (0.059)	0.095 (0.062)
ASO_nstem				0.029 (0.054)	-0.033 (0.064)
TSO_stem				0.036 (0.031)	0.033 (0.042)
TSO_nstem				-0.078* (0.045)	-0.016 (0.046)
BSO				-0.063* (0.035)	-0.079* (0.045)
4.rank_effort_math (vs1)					
ASO_stem				0.071 (0.059)	0.105* (0.063)
ASO_nstem				0.010 (0.056)	-0.031 (0.066)
TSO_stem				0.091** (0.036)	0.081* (0.047)
TSO_nstem				-0.088* (0.048)	-0.036 (0.050)
BSO				-0.084** (0.038)	-0.120** (0.048)
Observations	2913.000	2694.000	2694.000	2398.000	1609.000
R-squared					

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The picture does not change much.

4 School

In our sample, we have 2 ways to distinguish schools: type and cycle length.

Table 5: School Types and Cycle Lengths in the Sample (grade 2, level of obs: students)

Category	Frequency	Percent
School Type		
ASO-only	1,654	37.87%
AUT/MS (cycle 1 only)	900	20.61%
MLT/LAT (multilateral)	874	20.01%
TSO/BSO-only	939	21.50%
<i>Total</i>	4,367	100.00%
School Cycle Length		
Cycle 1 (grade 1 only)	1,628	36.82%
Cycle 1+2	96	2.17%
Cycle 1+2+3	2,516	56.90%
Cycle 1+2+3+4	182	4.12%
<i>Total</i>	4,422	100.00%

Schools in the sample differ along two institutional dimensions: the type of tracks they offer and the length of the educational cycle they provide. ASO-only schools offer exclusively academic tracks and therefore have a clear academic orientation. TSO/BSO-only schools provide only technical or vocational programs, so students entering these schools in grade 1–2 face substantially more limited upward mobility later on. Multilateral schools (MLT/LAT) offer the full range of tracks (ASO, TSO, BSO), allowing for more internal mobility. AUT/MS schools offer only the first cycle (grades 1–2), after which students must transition elsewhere; these schools account for about one-fifth of the sample.

Cycle length captures how many years of secondary education a school provides. Cycle 1 schools offer only grades 1–2, whereas Cycle 1+2+3 schools provide the full pathway up to the end of upper secondary. Cycle 1+2+3+4 schools are those that offer the extended professional programs associated with BSO tracks (students who do not transition to higher education).

Switching schools between grade 2 and grade 3. Because some schools only operate in cycle 1, we observe substantial mobility between grade 2 and grade 3. Nearly 50% of students in our sample switch schools at this transition point. As expected, switching is concentrated among students in AUT-type and cycle-1 schools, though we also observe mobility out of multilateral schools (and less so out of ASO-only or TSO/BSO-only schools).

Self-selection at grade 1. Students do not randomly choose schools: instead, there is strong self-selection along socioeconomic and ability lines. High-SES and higher-performing students disproportionately enter ASO-only schools offering longer academic cycles, while others concentrate in mixed or vocational schools with shorter programs. This early sorting is important because schools differ sharply in the advice they can give: ASO-only schools only recommend academic tracks, whereas TSO/BSO-only schools tend not to recommend ASO at all. Thus, the initial choice of school already shapes the vertical margin of tracking before teacher recommendations are issued. In contrast, horizontal differentiation (STEM vs. non-STEM within a track) appears much weaker, suggesting that school supply constraints/student self-selection operate primarily on the vertical dimension.

Table 6: Determinants of Sorting at grade 1 Into School Types (Marginal Effects)

	ASO-only	AUT/MS	MLT/ATL	TSO/BSO-only
Personality Traits				
Conscientiousness	0.169*** (0.005)	0.001 (0.004)	-0.045*** (0.004)	-0.125*** (0.003)
Agreeableness	-0.007 (0.005)	-0.010*** (0.004)	0.007* (0.004)	0.011*** (0.003)
Extraversion	-0.025*** (0.005)	0.005 (0.004)	0.013*** (0.004)	0.007** (0.003)
Ability (IQ Indicator)				
Male	-0.043*** (0.008)	0.051*** (0.007)	-0.067*** (0.007)	0.059*** (0.006)
Socioeconomic Status				
High SES	0.147*** (0.009)	-0.026*** (0.008)	-0.010 (0.008)	-0.112*** (0.007)
Observations	14,665			

Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 7: Determinants of Sorting Into School Cycle Length (Marginal Effects)

	Cycle 1	Cycle 1+2	Cycle 1+2+3	Cycle 1+2+3+4
Personality Traits				
Conscientiousness	0.023*** (0.005)	-0.007*** (0.002)	0.010* (0.005)	-0.025*** (0.002)
Agreeableness	-0.007 (0.005)	0.000 (0.001)	0.003 (0.005)	0.004** (0.002)
Extraversion	0.003 (0.005)	0.003** (0.001)	-0.009* (0.005)	0.003* (0.002)
Ability (IQ Indicator)				
Male	0.016** (0.008)	0.006** (0.003)	0.052*** (0.008)	-0.074*** (0.003)
Socioeconomic Status				
High SES	0.010 (0.009)	-0.006** (0.003)	0.030*** (0.009)	-0.034*** (0.003)
Observations	14,830			

Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

5 Conclusion

Across all analyses, three patterns consistently emerge. First, A/B/C certificates are driven overwhelmingly by performance and teacher-based rankings, with C-certificates remaining difficult to predict due to low probability and potential idiosyncratic shocks. Second, conditional on receiving an A-certificate, teacher advice systematically reflects relative performance within the classroom, gender norms, and the institutional constraints of the school. Third, students sort strongly and early into school types and cycle lengths, and this initial sorting already shapes the vertical tracking margin before teachers intervene. These findings motivate investigating more the supply side (Schools) at the vertical and horizontal level.

6 Appendix

Table 8: Determinants of C-Certificate Decisions (1) vs A certificates (0)

	(1)	(2)	(3)	(4)
<i>Student characteristics</i>				
Male (ref: female)	0.009 (0.008)	0.011 (0.007)	0.009 (0.007)	0.008 (0.007)
Numeric IQ (std.)	-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Verbal IQ (std.)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.000 (0.000)
Conscientiousness (std.)	-0.022*** (0.006)	-0.009 (0.006)	-0.010* (0.006)	-0.010* (0.005)
Agreeableness (std.)	-0.009* (0.005)	-0.011** (0.005)	-0.003 (0.005)	-0.003 (0.005)
Extraversion (std.)	0.002	0.001	0.000	0.003

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	(1)	(2)	(3)	(4)
	(0.004)	(0.004)	(0.004)	(0.003)
<i>Parental education (ref: low)</i>				
Medium parental education	-0.017** (0.008)	-0.022*** (0.007)	-0.017** (0.007)	-0.009 (0.006)
High parental education	-0.007 (0.007)	-0.014** (0.006)	-0.008 (0.005)	-0.001 (0.005)
<i>Standardized grades</i>				
Standardized Dutch grade	-0.012** (0.005)	-0.005 (0.004)	-0.004 (0.004)	-0.003 (0.004)
Standardized math grade	-0.000 (0.004)	0.000 (0.004)	0.001 (0.004)	-0.001 (0.004)
<i>Track (ref: BSO)</i>				
TSO	0.015 (0.021)	-0.015 (0.019)	-0.009 (0.019)	0.011 (0.019)
ASO	0.030 (0.020)	-0.007 (0.019)	0.004 (0.018)	0.020 (0.019)
<i>Teacher-reported math rank (ref: lowest quintile)</i>				
Math rank Q2 (teacher)		-0.106*** (0.022)	-0.097*** (0.021)	-0.069*** (0.018)
Math rank Q3 (teacher)		-0.128*** (0.028)	-0.109*** (0.027)	-0.079*** (0.023)
Math rank Q4 (teacher)		-0.128*** (0.027)	-0.103*** (0.024)	-0.076*** (0.022)
Math rank Q5 (teacher)		-0.128*** (0.027)	-0.095*** (0.024)	-0.078*** (0.022)
<i>Teacher-reported Dutch rank (ref: lowest quintile)</i>				
Dutch rank Q2 (teacher)		-0.114*** (0.028)	-0.107*** (0.025)	-0.077*** (0.024)
Dutch rank Q3 (teacher)		-0.106*** (0.024)	-0.091*** (0.022)	-0.073*** (0.021)
Dutch rank Q4 (teacher)		-0.109*** (0.026)	-0.085*** (0.023)	-0.070*** (0.022)
Dutch rank Q5 (teacher)		-0.099*** (0.025)	-0.065*** (0.023)	-0.069*** (0.020)
<i>Teacher assessments</i>				
General effort (std., teacher)			-0.033*** (0.010)	
Parental involvement (std.)			-0.012** (0.005)	-0.014*** (0.005)
<i>Subject-specific effort (ref: very low/low = level 1)</i>				
Dutch effort level 2				-0.043** (0.020)
Dutch effort level 3				-0.044** (0.017)
Dutch effort level 4				-0.036* (0.019)
Math effort level 2				-0.138*** (0.032)
Math effort level 3				-0.162*** (0.033)
Math effort level 4				-0.152*** (0.033)
Constant	0.210*** (0.050)	0.360*** (0.061)	0.456*** (0.081)	0.426*** (0.065)

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	(1)	(2)	(3)	(4)
Observations	2925	2647	2573	2574
R-squared	0.057	0.139	0.160	0.236
School FE	No	No	No	Yes

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: Determinants of B/C-Certificate Decisions (1) vs A certificate (0)

	(1)	(2)	(3)	(4)
<i>Student characteristics</i>				
Male (ref: female)	0.008 (0.015)	0.013 (0.014)	0.008 (0.015)	-0.008 (0.015)
Numeric IQ (std.)	-0.002*** (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)
Verbal IQ (std.)	0.000 (0.001)	0.001* (0.001)	0.001 (0.001)	0.001 (0.001)
Conscientiousness (std.)	-0.074*** (0.010)	-0.020** (0.009)	-0.023** (0.010)	-0.037*** (0.010)
Agreeableness (std.)	-0.012** (0.006)	-0.009 (0.005)	0.004 (0.005)	0.007 (0.005)
Extraversion (std.)	0.015* (0.007)	0.002 (0.007)	0.003 (0.008)	0.011 (0.008)
<i>Parental education (ref: low education)</i>				
Medium parental education	-0.029* (0.016)	-0.037** (0.015)	-0.034** (0.015)	-0.027* (0.014)
High parental education	-0.045*** (0.015)	-0.045*** (0.014)	-0.043*** (0.014)	-0.032** (0.013)
<i>Standardized grades</i>				
Standardized Dutch grade	-0.030** (0.012)	0.001 (0.010)	0.001 (0.010)	-0.001 (0.008)
Standardized math grade	-0.021*** (0.007)	-0.017** (0.006)	-0.017*** (0.006)	-0.016** (0.006)
<i>Track (ref: BSO)</i>				
TSO	0.395*** (0.058)	0.213*** (0.047)	0.226*** (0.047)	0.224*** (0.035)
ASO	0.329*** (0.027)	0.140*** (0.027)	0.163*** (0.026)	0.162*** (0.030)
<i>Teacher-reported math rank (ref: Q1 = lowest quintile)</i>				
Math rank Q2		-0.139*** (0.026)	-0.124*** (0.026)	-0.117*** (0.026)
Math rank Q3		-0.252*** (0.029)	-0.220*** (0.027)	-0.212*** (0.027)
Math rank Q4		-0.260*** (0.026)	-0.218*** (0.023)	-0.204*** (0.025)
Math rank Q5		-0.269*** (0.027)	-0.210*** (0.023)	-0.196*** (0.024)
<i>Teacher-reported Dutch rank (ref: Q1)</i>				
Dutch rank Q2		-0.133*** (0.036)	-0.115*** (0.036)	-0.121*** (0.033)
Dutch rank Q3		-0.293*** (0.027)	-0.259*** (0.027)	-0.255*** (0.028)
Dutch rank Q4		-0.334*** (0.030)	-0.287*** (0.033)	-0.291*** (0.031)
Dutch rank Q5		-0.318*** (0.030)	-0.255*** (0.033)	-0.278*** (0.031)

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	(1)	(2)	(3)	(4)
		(0.029)	(0.035)	(0.029)
<i>Teacher assessments</i>				
General effort (std.)			-0.068*** (0.013)	
Parental involvement (std.)			0.000 (0.008)	-0.015** (0.007)
<i>Subject-specific effort (ref: level 1 = lowest effort)</i>				
Dutch effort level 2				-0.014 (0.027)
Dutch effort level 3				-0.027 (0.027)
Dutch effort level 4				-0.016 (0.023)
Math effort level 2				-0.117*** (0.031)
Math effort level 3				-0.184*** (0.031)
Math effort level 4				-0.202*** (0.032)
<i>Constant</i>				
Constant	0.380*** (0.077)	0.579*** (0.068)	0.711*** (0.074)	0.747*** (0.082)
Observations	3426	3103	3013	3014
R-squared	0.159	0.334	0.340	0.396
School FE	No	No	No	Yes

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Determinants of B vs C Certificate

	(1)	(2)	(3)	(4)
<i>Student characteristics</i>				
Male (ref: female)	0.040 (0.039)	0.041 (0.042)	0.040 (0.043)	0.038 (0.034)
Numeric IQ (std.)	0.000 (0.001)	0.000 (0.001)	0.001 (0.001)	0.001 (0.002)
Verbal IQ (std.)	-0.002* (0.001)	-0.003* (0.002)	-0.003* (0.002)	-0.001 (0.001)
Conscientiousness (std.)	-0.017 (0.019)	-0.006 (0.019)	-0.006 (0.019)	-0.007 (0.021)
Agreeableness (std.)	-0.011 (0.015)	-0.011 (0.017)	-0.002 (0.017)	0.004 (0.017)
Extraversion (std.)	-0.003 (0.014)	-0.004 (0.013)	-0.008 (0.013)	-0.007 (0.014)
<i>Parental education (ref: low)</i>				
Medium parental education	-0.049* (0.025)	-0.051* (0.026)	-0.038 (0.025)	0.000 (0.028)
High parental education	-0.010 (0.038)	-0.025 (0.037)	-0.007 (0.041)	-0.020 (0.040)
<i>Standardized grades</i>				
Standardized Dutch grade	-0.010 (0.011)	-0.004 (0.010)	-0.000 (0.011)	-0.006 (0.009)
Standardized math grade	0.006 (0.007)	0.010 (0.007)	0.008 (0.007)	0.012 (0.008)

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	(1)	(2)	(3)	(4)
<i>Track (ref: BSO)</i>				
TSO	-0.812*** (0.065)	-0.819*** (0.075)	-0.787*** (0.080)	-0.603*** (0.091)
ASO	-0.772*** (0.064)	-0.802*** (0.078)	-0.774*** (0.083)	-0.647*** (0.089)
<i>Teacher-reported math rank (ref: lowest quintile)</i>				
Math rank Q2		-0.061** (0.027)	-0.052* (0.026)	-0.057* (0.030)
Math rank Q3		-0.104** (0.042)	-0.090** (0.042)	-0.095** (0.047)
Math rank Q4		-0.090* (0.047)	-0.071 (0.045)	-0.106** (0.047)
Math rank Q5		-0.078 (0.048)	-0.040 (0.059)	0.061 (0.092)
<i>Teacher-reported Dutch rank (ref: lowest quintile)</i>				
Dutch rank Q2		-0.057 (0.040)	-0.048 (0.038)	-0.044 (0.038)
Dutch rank Q3		-0.003 (0.039)	0.009 (0.036)	-0.004 (0.039)
Dutch rank Q4		0.023 (0.073)	0.050 (0.075)	0.017 (0.057)
Dutch rank Q5		0.077 (0.131)	0.114 (0.129)	0.056 (0.093)
<i>Teacher assessments</i>				
General effort (std.)			-0.029 (0.020)	
Parental involvement (std.)			-0.026* (0.016)	-0.013 (0.016)
<i>Subject-specific effort (ref: lowest effort)</i>				
Dutch effort level 2				-0.064 (0.053)
Dutch effort level 3				-0.031 (0.048)
Dutch effort level 4				-0.025 (0.071)
Math effort level 2				-0.021 (0.026)
Math effort level 3				-0.072* (0.042)
Math effort level 4				-0.104* (0.061)
<i>Constant</i>				
Constant	1.242*** (0.121)	1.272*** (0.135)	1.335*** (0.151)	1.008*** (0.172)
Observations	593	538	520	520
R-squared	0.403	0.424	0.446	0.611
School FE	No	No	No	Yes

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$